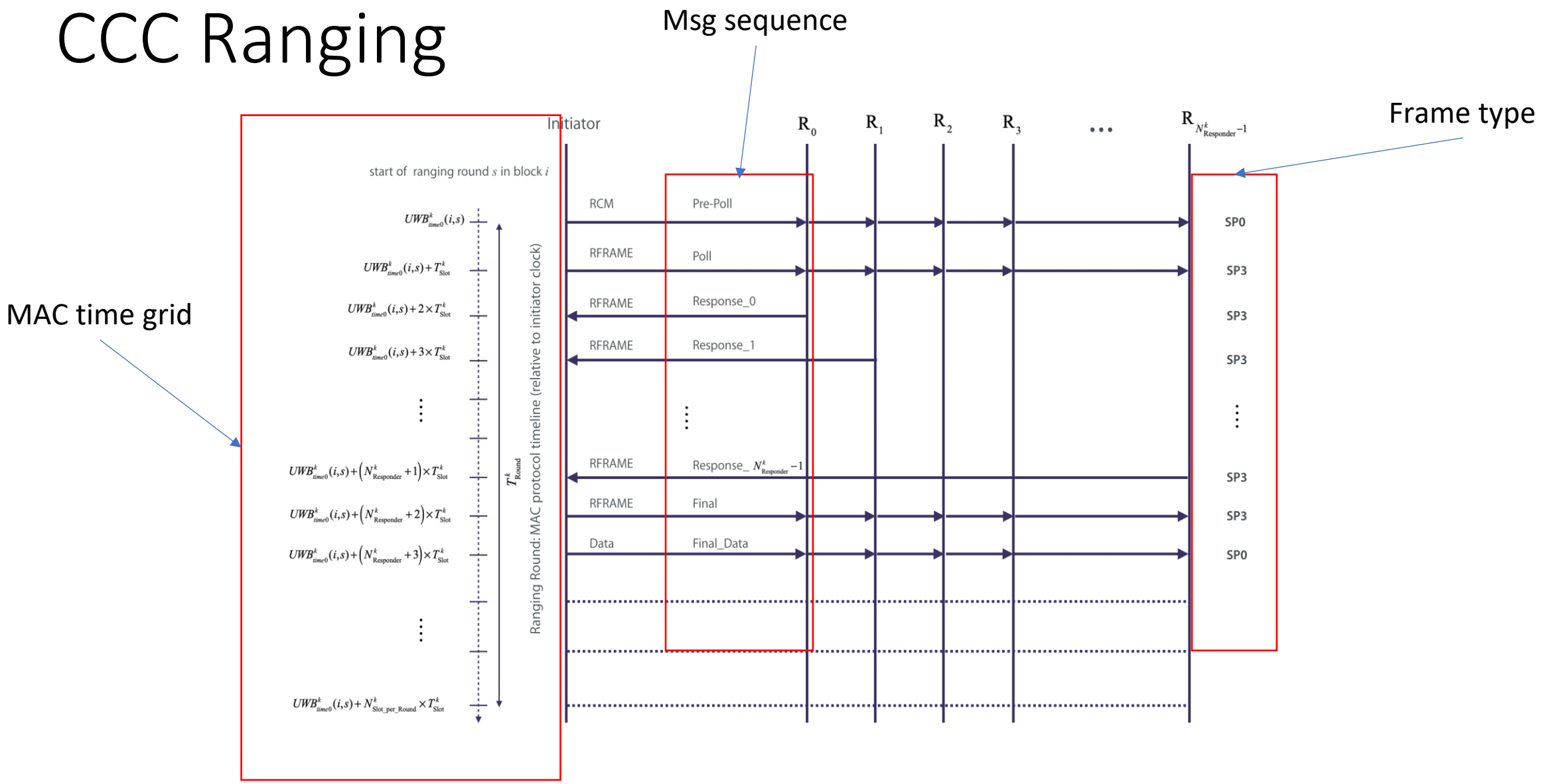
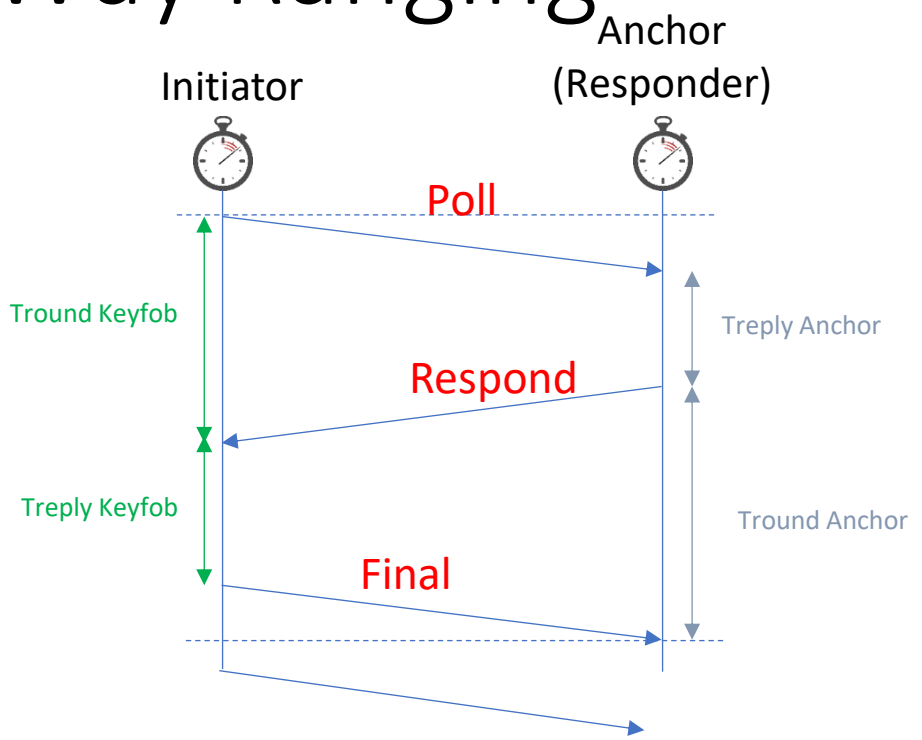
CCC Ranging


Msg sequence
Frame type
MAC time grid
Initiator
R_0
R_1
R_2
R_3
$R_{N^k_{Responder}-1}$
start of ranging round s in block i
$UWB^k_{time0}(i,s)$
$UWB^k_{time0}(i,s)+T^k_{Slot}$
$UWB^k_{time0}(i,s)+2\times T^k_{Slot}$
$UWB^k_{time0}(i,s)+3\times T^k_{Slot}$
$UWB^k_{time0}(i,s)+(N^k_{Responder}+1)\times T^k_{Slot}$
$UWB^k_{time0}(i,s)+(N^k_{Responder}+2)\times T^k_{Slot}$
$UWB^k_{time0}(i,s)+(N^k_{Responder}+3)\times T^k_{Slot}$
$UWB^k_{time0}(i,s)+N^k_{Slot_per_Round}\times T^k_{Slot}$
T^k_{Round}
Ranging Round: MAC protocol timeline (relative to initiator clock)
RCM
Pre-Poll
SP0
RFRAME
Poll
SP3
RFRAME
Response_0
SP3
RFRAME
Response_1
SP3
RFRAME
Response_ $N^k_{Responder}-1$
SP3
RFRAME
Final
SP3
Data
Final_Data
SP0

Double Sided - Two Way Ranging

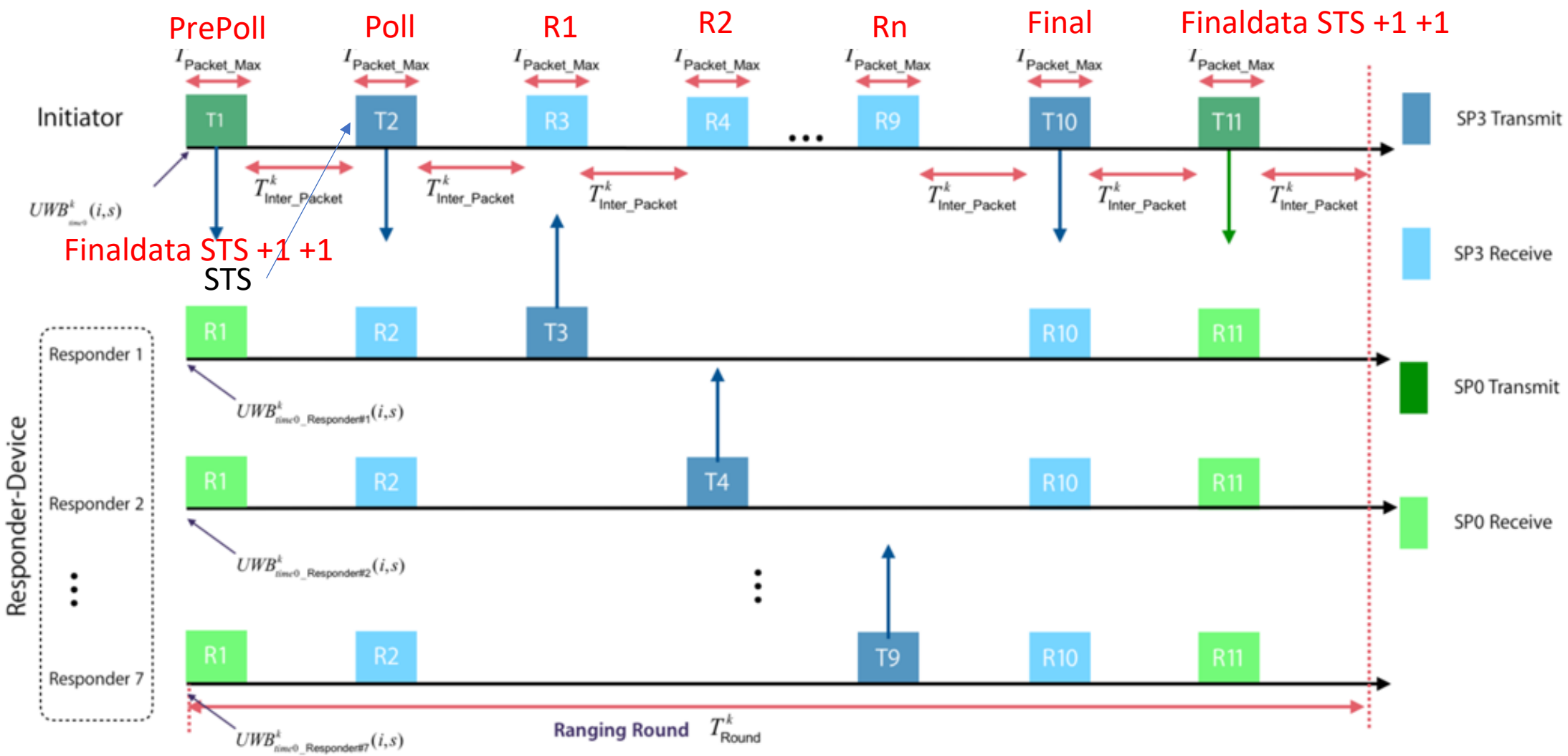
- Consists of 3 messages:
  - Ranging request
  - Ranging Response
  - Final message (with anchor timestamps)

$$ToF = \frac{T_{roundA} \times T_{roundK} - T_{replyA} \times T_{replyK}}{t_{roundA} + t_{roundK} + T_{replyA} + T_{replyK}}$$



+ Automatic compensation for clock differences in the two devices
+ Time computation can be done on the car
+ More secure (preamble injection more difficult)
– need more current

MESSAGE SEQUENCE



MESSAGE SEQUENCE

- T_{Chap}: Unit of time for the MAC protocol. All durations (except for T_{Packet_Max} defined below) of the MAC protocol are integer multiples of T_{Chap}.

$$T_{Chap} = \frac{1}{3}\text{ms} = 400\ \text{RSTU}$$

where the RSTU is defined in [33] as $416/(499.2\ \text{MHz}) \cup 833.33$ ns.

- T_{Block_Min} : Minimum ranging block duration, which is set to 96ms, is an integer multiple of T_{Chap}

$$T(\text{Block_Min}) = 288 \times T_\text{Chap} = 96\text{ms}$$

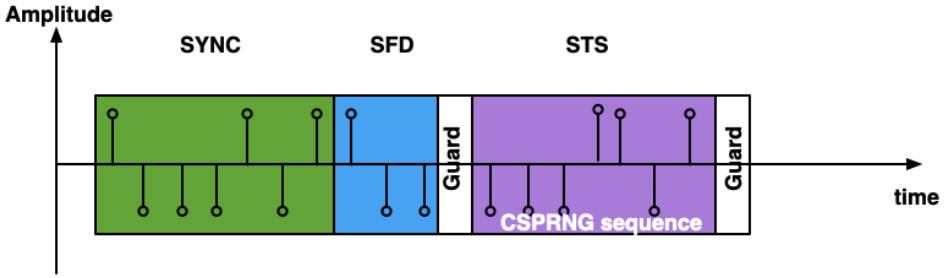
Each block is divided into a number of slots.

$$T^{k}_{\text{Slot}} = N^{k}_{\text{Chap_per_Slot}} \times T_{\text{Chap}}$$

$$T^{k}_{\text{Round}} = N^{k}_{\text{Slot_per_round}} \times T^{k}_{\text{Slot}} = N^{k}_{\text{Slot_per_round}} \times N^{k}_{\text{Chap_per_Slot}} \times T_{\text{Chap}}$$

Ranging Frames

Messages:
Poll
Response
Final



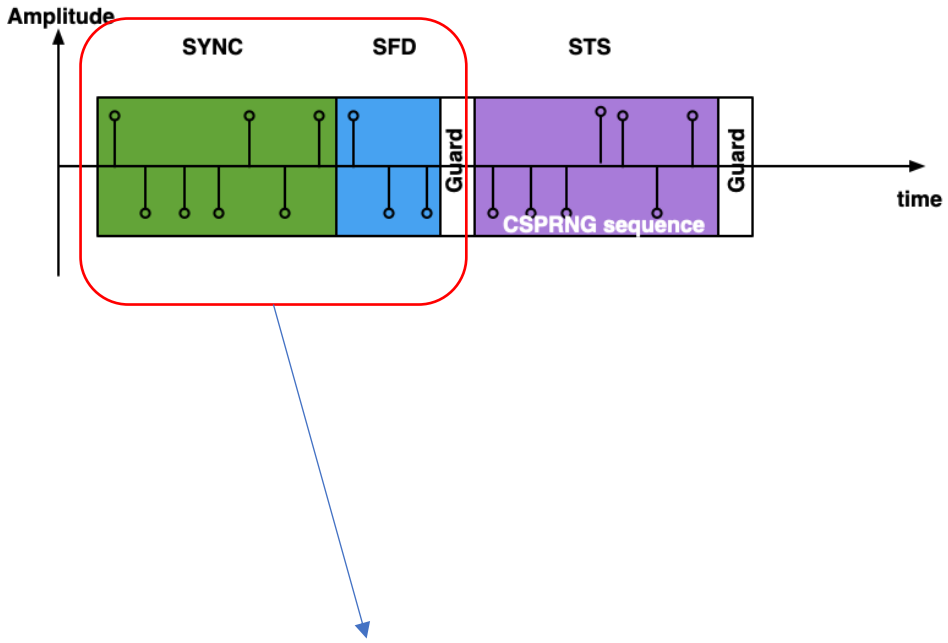
What to do :
- Check error
- Take time stamp (important)
- incriment STS

If error :
- ,could continue because we only need IRQ for timestamp

Meaning of SP3 frame:
- only needed for timestamp
- nessesary for TOF calculation

Ranging Frames

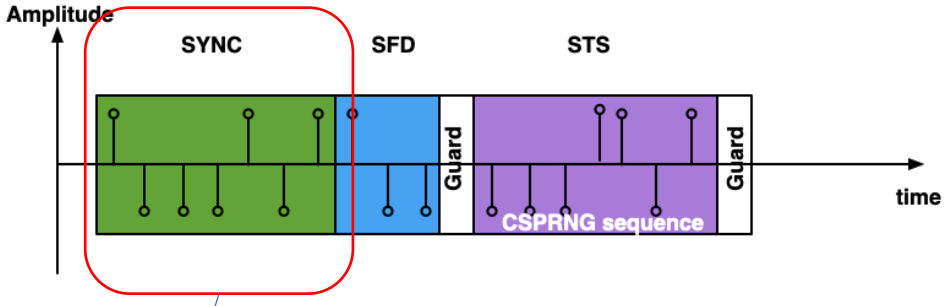
Messages:
Poll
Response
Final



Only for syncing purpose (by HW)

Ranging Frames

Messages:
Poll
Response
Final



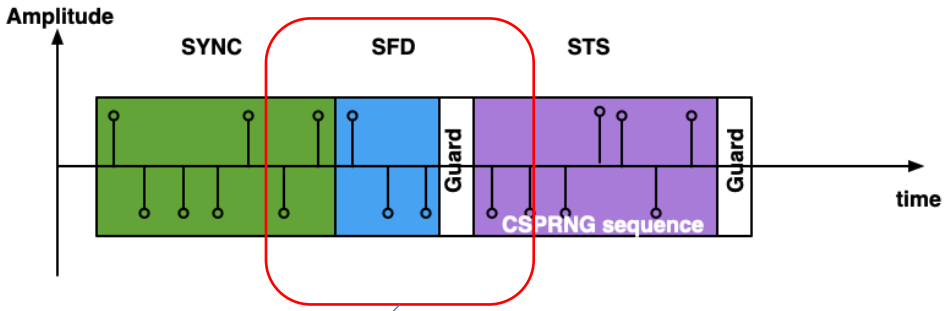
16

Table 21-2: The Mandatory Length-127 Ternary Code Sequences.

Code index	Ternary Code Sequence	UWB channels
9	+00+000-0--00--+0+0+00-+-++0+0000++-000+00-00--0-+0+0--0-+++0++000+-0+00-0++-0+++00-+00+0+0-0++-+--+000000+00000-+0000-0-000--+	5,9
10	++00+0-+00+00+000000-000-00--000-0+-+0-0+-0-+00000+-00++0-0+00--+00++-+0+-0+0000-0-0-0-++-+0+00+0+000-+0+++000----+++0000+++0--	5,9
11	-+-0000+00--00000-0+0+0+-0+00+00+0-00-+++00+000-+0+0-0000+++++-+0+--0+-0++--0-000+0-+00+0+----000-000000-+00+-0++000++-00++-0-0	5,9
12	-+0++000000-0+0-+0---+-++00-+0++0+0+0+000-00-00-+00+-++000-+-0-++0-0++++0-00-0++00+0+00++-00+000+-000-0--+0000-0000--0+00000+--	5,9

Ranging Frames

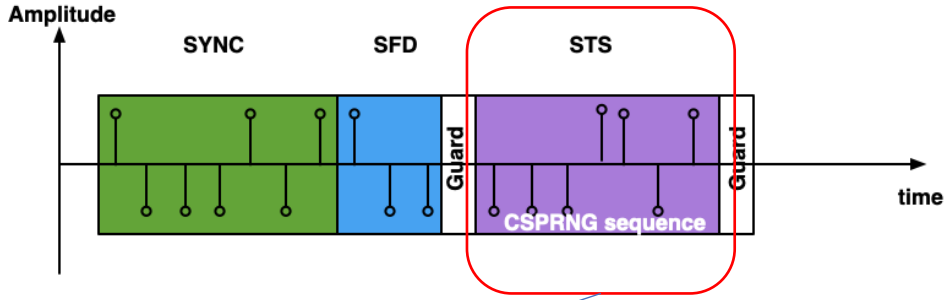
Messages:
Poll
Response
Final



Establish timing 8 bit

Ranging Frames

Messages:
Poll
Response
Final



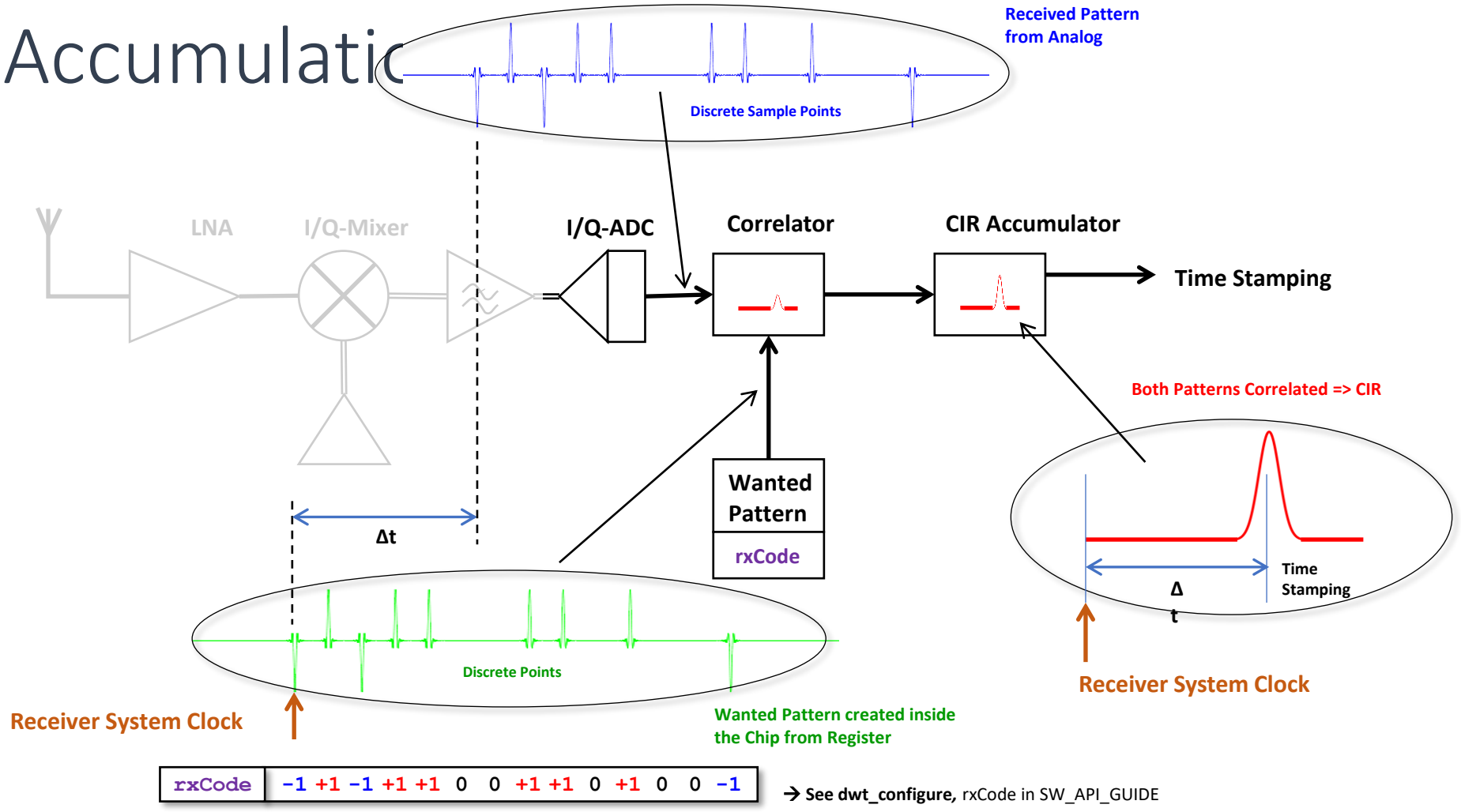
21.4.3 *Scrambled Timestamp Sequence (STS)*

UWB PHY supports Secure Ranging mechanisms to prevent different forms of attacks carried out by a "rogue" device injecting signal energies with the objective of making ranging receivers misread the distance between associated ranging peers. Scrambled Timestamp Sequence (STS) provides an AES-generated ranging waveform to protect against range measurement spoofing. DRBG based STS is resilient to various types of distance manipulation attacks like Brute force attack, Cicada attack, preamble injection attack and preamble EDLC. Details on how STS sequence is generated is specified in [33]. The number of bits used for STS is specified in Table 21-1. STS bits are mapped to pulses as follows:

1. Bit 1 is mapped to a negative pulse
2. Bit 0 is mapped to a positive pulse

In the BPRF (64 MHz PRF) mode, there is one pulse every 8 chips (spreading = 8)

Accumulatic

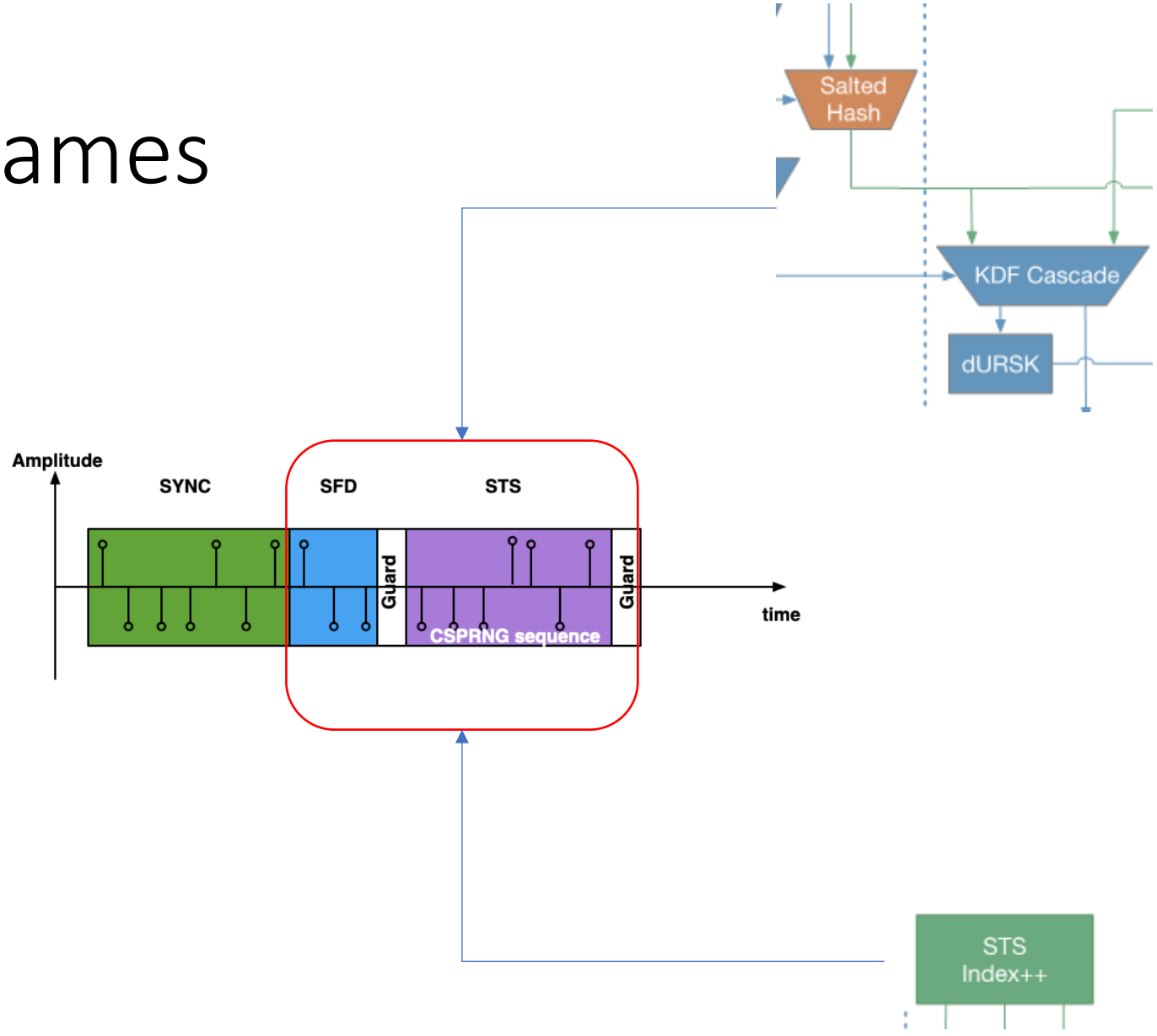


rxCode	-1	+1	-1	+1	+1	0	0	+1	+1	0	+1	0	0	-1

→ **See dwt_configure,** rxCode in SW_API_GUIDE

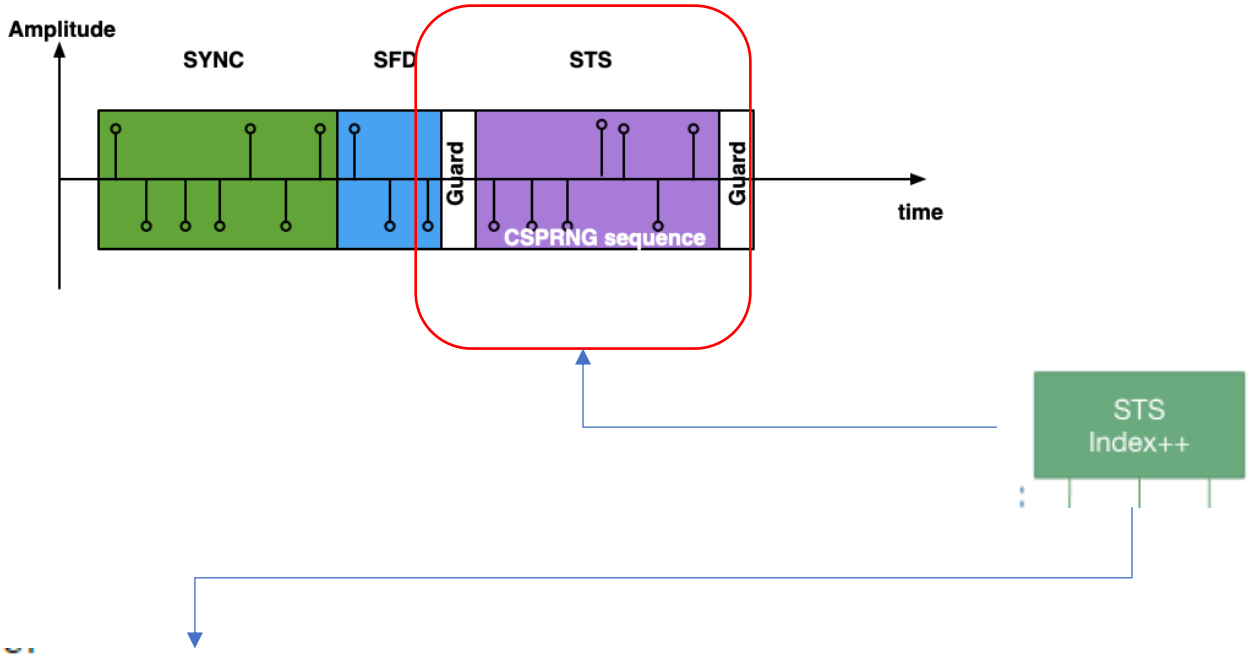
Ranging Frames

Messages:
Poll
Response
Final



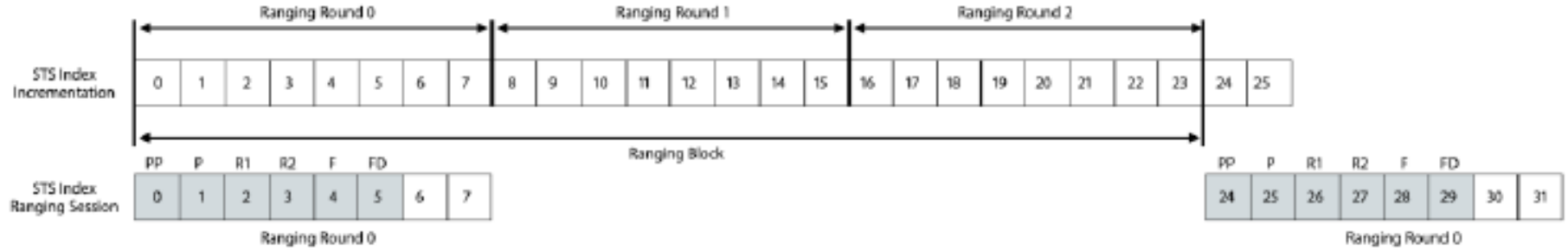
Ranging Frames

Messages:
Poll
Response
Final

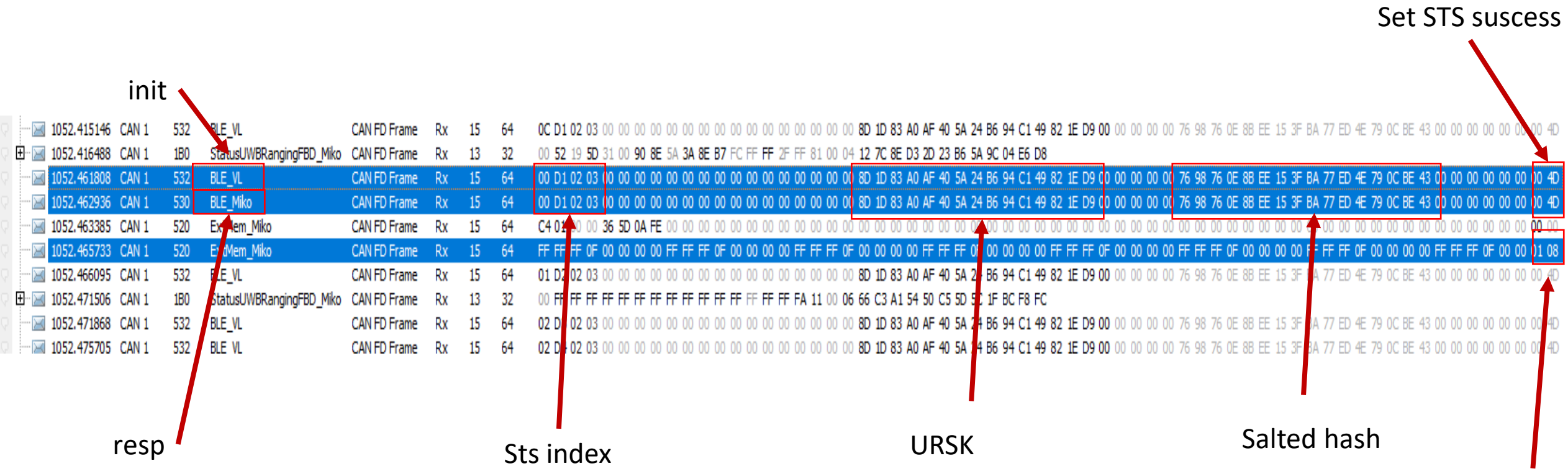


3

Figure 20-8: Basic Principles of STS Incrementation.



4


Set STS suscess
init
resp
Sts index
URSK
Salted hash
Sts miss match error

Data Frames

For payload messages used:

- Pre-Poll
- Final Data

Figure 20-7: SP0 Packet Content.

MHR (23 Octets)	MAC Payload Frame Payload (variable) + MIC (8 octects)	MFR (2 Octets)

20.5.4.1 MHR Field



Bits: 0–1	2–8	9	10	11–12	13–18
Data Rate	Frame Length	Ranging	Reserved	Preamble Duration	SECDED

Data Frames

For payload messages used:

- Pre-Poll
- Final Data

Figure 20-7: SP0 Packet Content.

MHR (23 Octets)	MAC Payload Frame Payload (variable) + MIC (8 octects)	MFR (2 Octets)

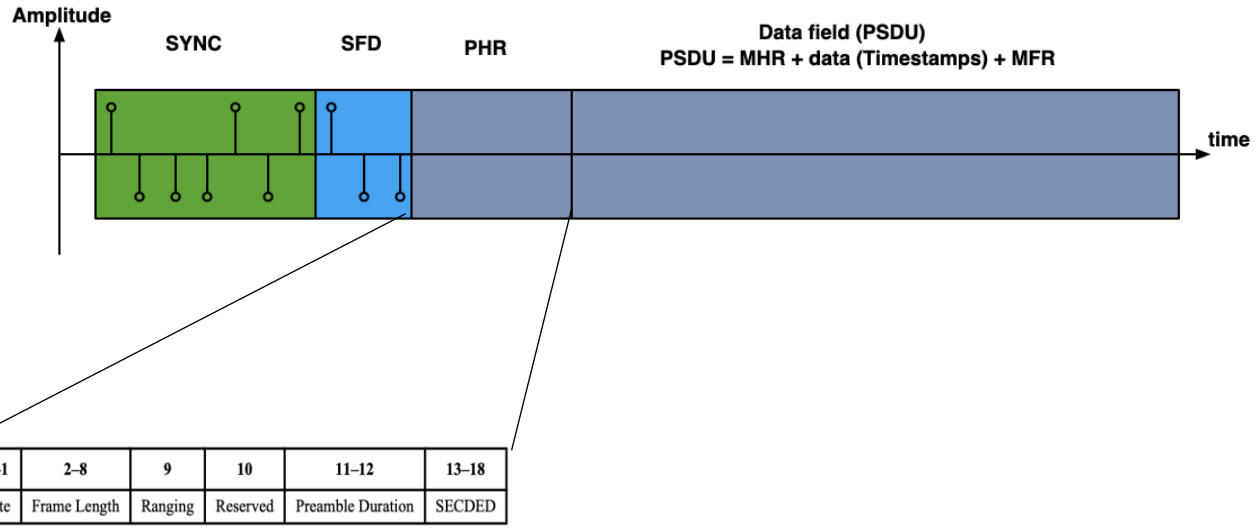
20.5.4.1 MHR Field



Bits: 0–1	2–8	9	10	11–12	13–18
Data Rate	Frame Length	Ranging	Reserved	Preamble Duration	SECDED

Data Frames

For payload messages used:
- Pre-Poll



Bits: 0–1	2–8	9	10	11–12	13–18
Data Rate	Frame Length	Ranging	Reserved	Preamble Duration	SECDED

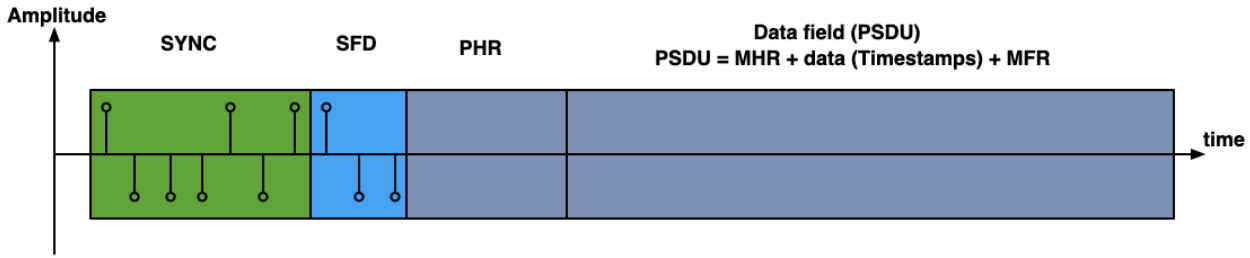
What to do :
- Check error
- Decrypt
- get first path info
- get Sts index for POLL

If error :
- calculate time for next round

Meaning of Prepoll:
- only needed for frame syncing
- not nessesary for TOF calculation

Data Frames

For payload messages used:
- Pre-Poll



1

Table 20-3: Pre-Poll Request Message and its parameters.

UWB MAC Message	UWB MAC Message ID	Parameters
Pre-POLL	1	UWB_Session_ID, Poll_STS_Index, Ranging_Block, Hop_Flag, Round_Index

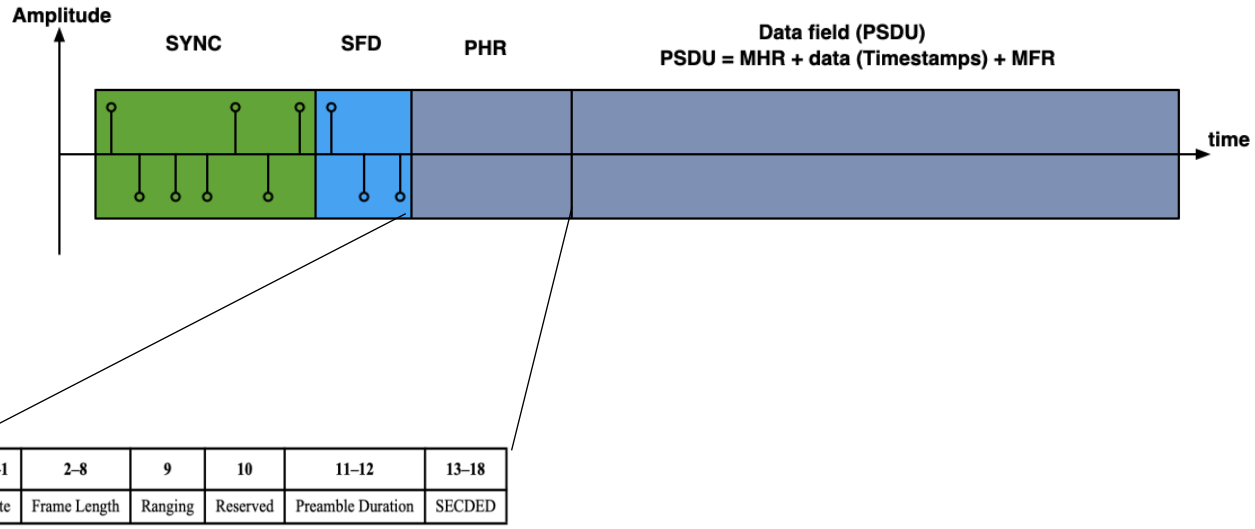
2

Table 20-4: The definition of parameters in the Pre-Poll Message.

Parameters	Length (bytes)	Value	Description
UWB_Session_ID	4	0 - (2^{32}-1)	ID of the UWB ranging session
Poll_STS_Index	4	0 - (2^{31}-1)	STS index of the succeeding POLL message
Ranging_Block	2	0 - 65535	Session's Index of current ranging block
Hop_Flag	1	0:No Hopping 1:Hopping	Hop flag for current ranging block as set from the ranging exchange in the previous ranging block For no hopping configuration this field is always 0 For continuous hopping configuration this field is always 1 (except for the first ranging block with index 0 after start or recovery of a ranging session).
Round_Index	2	0-65535	The ranging round index for the current ranging block as set from the ranging exchange in the previous ranging block. This field is set to 0 in first ranging block where we always start in ranging round 0.

Data Frames

For payload messages used:
- Final



Bits: 0–1	2–8	9	10	11–12	13–18
Data Rate	Frame Length	Ranging	Reserved	Preamble Duration	SECDED

What to do :
- Check error
- Decrypt
- get first path info
- get Sts index for preceding Final

If error :
- calculate time for next round

Meaning of Prepoll:
- needed for TOF . Contain all nessesary data for previous sequence

Data Frames

For payload messages used:
- Final

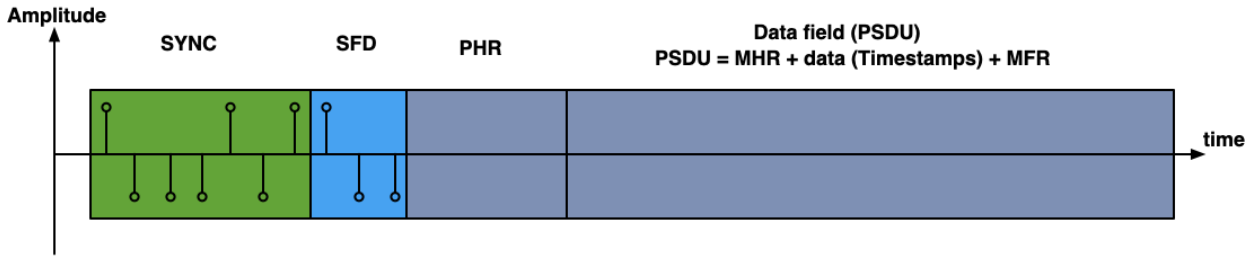


Table 20-5: Final_Data Message and its parameters.

UWB MAC Message	UWB MAC Message ID	Parameters
Final_Data	2	UWB_Session_ID, Ranging_Block, Hop_Flag, Round_Index, Final_STS_Index,
		Ranging_Timestamp_FINAL_TX, Number_Ranging_Responders, Responder_Index, Ranging_Timestamp_Responder_*l*, Ranging_Timestamp_Uncertainty_Responder_*l*, Ranging_Status_Responder_*l*, …

Data Frames

For payload messages used:
- Final

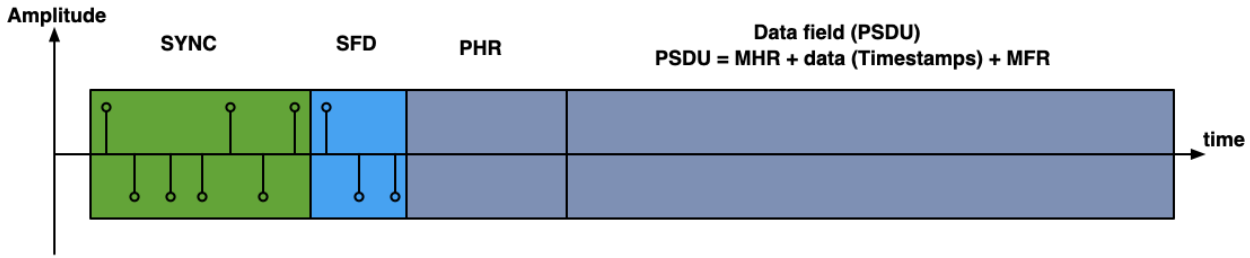


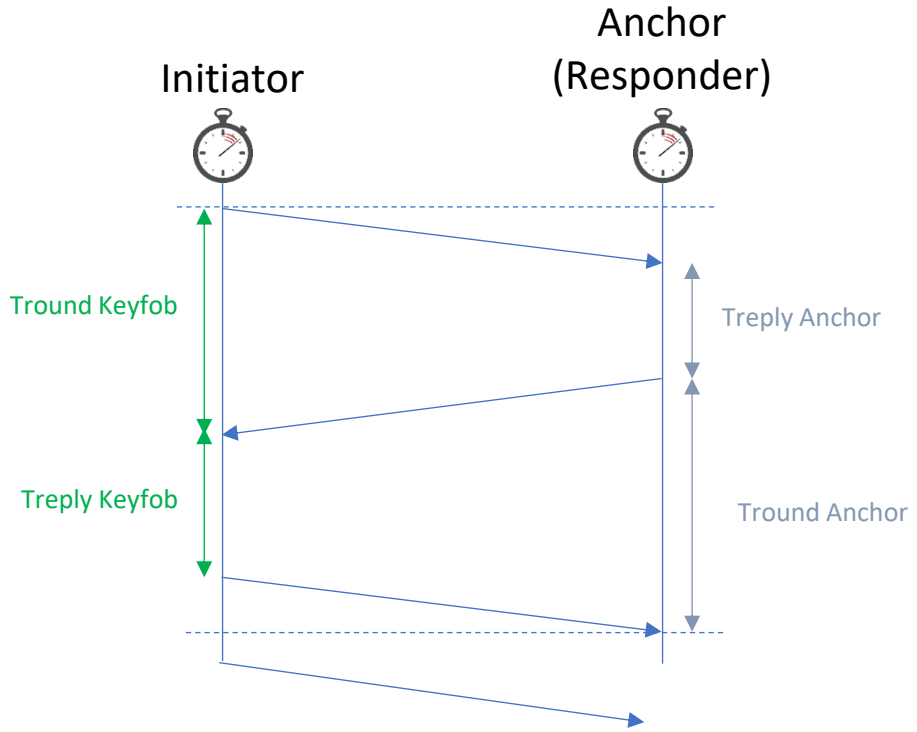
Table 20-6: The definition of parameters in the Final_Data Message.

Parameters	Length (bytes)	Value	Description
UWB_Session_ID	4	0–(2^{32}-1)	ID of the UWB ranging session
Ranging_Block	2	0–65535	Index of current ranging Block
Hop_Flag	1	0: No Hopping 1: Hopping	Hop Flag for next ranging Block For no hopping configuration this field is always 0 For continuous hopping configuration this field is always 1
Round_Index	2	0 – 65535	The ranging round in which the ranging cycle will be executed in the next ranging block
Final_STS_Index	4	0 – (2^{31}-1)	STS index of the preceding Final message.
Ranging_Timestamp_FINAL_TX	4	Range = 0 to 0XFFFFFFFF Timestamp = N × 1/128 × 1/499.2e6 sec Time Range = 0 to 67.21 ms	Time difference between POLL and Final messages transmit times at the initiator.
Number_Ranging_Responders	1	0–255	Number of timestamps to follow in this message
Responder_Index	1	0–255	Index of Responder *l* whose timestamp data is referred to in the Ranging_Timestamp_Responder_*l*, Ranging_Timestamp_Uncertainty_Responder_-*l*, and Ranging_Status_Responder_*l* fields
Ranging_Timestamp_Responder_*l*	4	Range = 0 to 0XFFFFFFFF Timestamp=N × 1/128 × 1/499.2e6 sec Time Range = 0 to 67.21 ms	Time difference between POLL and RESPONSE of Responder *l*, as received by the initiator.
Ranging_Timestamp_Uncertainty_-Responder_*l*	1	See Section 6.9.1.7 in [33]	Range of values from 1.5 cm–3.6 m at various confidences
Ranging_Status_Responder_*l*	1	See Table 20-7	The status for the response frame from the responder. See the corresponding description for the ranging status in Table 20-7
…	…	…	Repeat parameters Responder_Index to Ranging_Status_Responder_*l* for each responder whose time stamp data has been added (up to Number_Ranging_Responders times)

B2B – SELF TEST FEATURE

- Consists of 3 messages:
  - Ranging request
  - Ranging Response
  - Final message (with anchor timestamps)

$$ToF = \frac{T_{roundA} \times T_{roundK} - T_{replyA} \times T_{replyK}}{t_{roundA} + t_{roundK} + T_{replyA} + T_{replyK}}$$

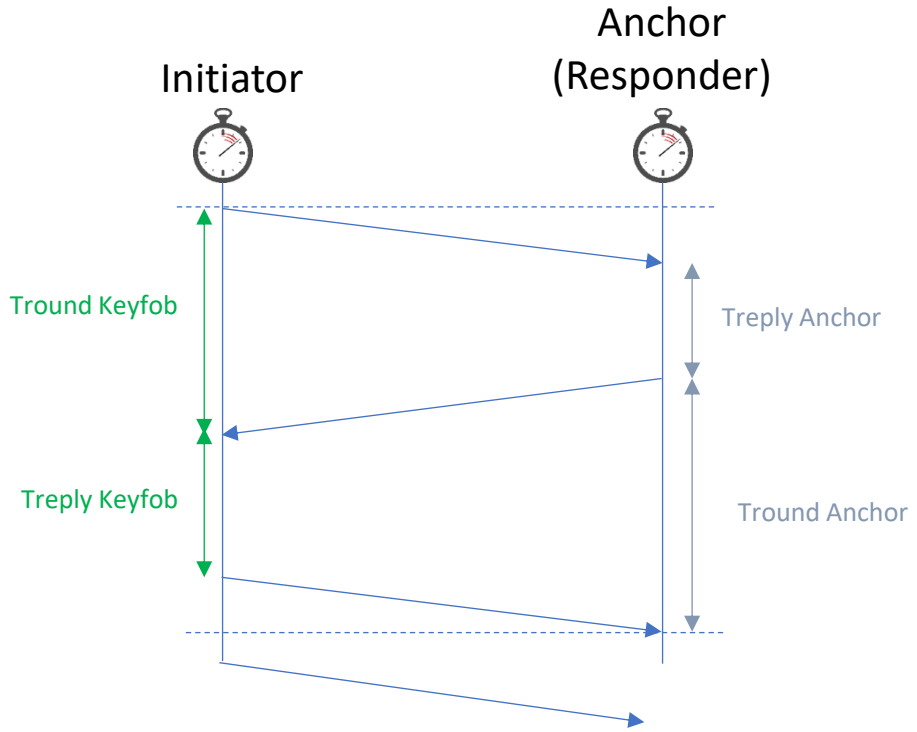


+ Automatic compensation for clock differences in the two devices
+ Time computation can be done on the car
+ More secure (preamble injection more difficult)
– need more current

B2B – SELF TEST FEATURE

- Consists of 3 messages:
  - Ranging request
  - Ranging Response
  - Final message (with anchor timestamps)

$$ToF = \frac{T_{roundA} \times T_{roundK} - T_{replyA} \times T_{replyK}}{t_{roundA} + t_{roundK} + T_{replyA} + T_{replyK}}$$



+ Automatic compensation for clock differences in the two devices
+ Time computation can be done on the car
+ More secure (preamble injection more difficult)
– need more current

B2B – SELF TEST FEATURE

Perform as CCC sequence:

- 1 board will be Initiator
- The rest will be Responder

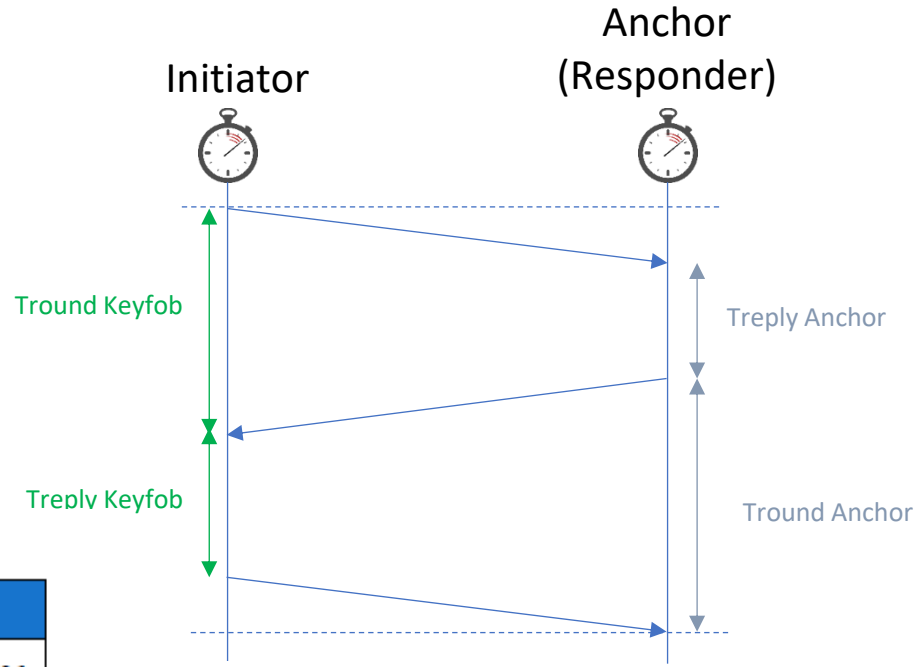


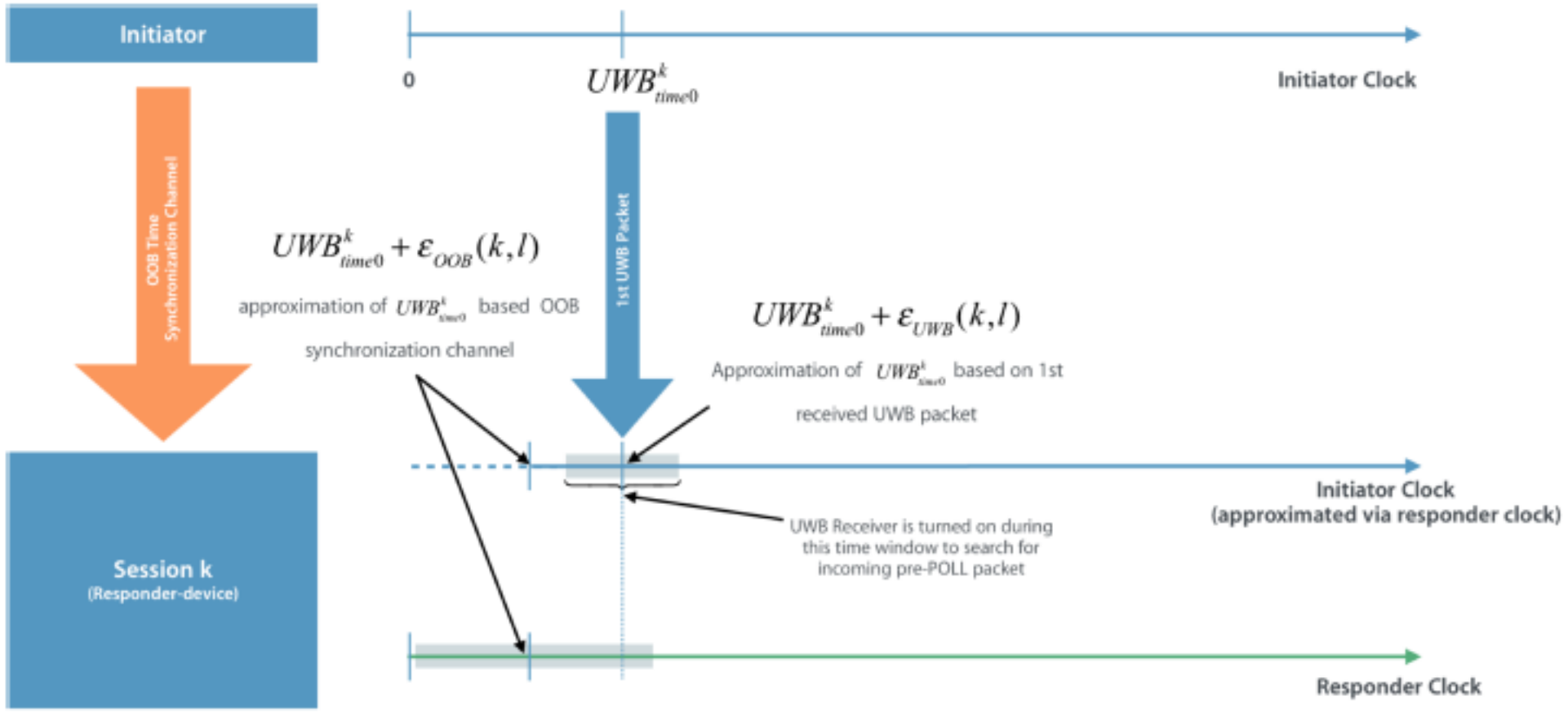
Table G-1: Number of Valid Ranging Rounds per 96 ms.

Number of Slots (per 96 ms)		$N^k_{Slot_per_Round}$											
$N^k_{Chap_per_Slot}$	Resulting T^k_{Slot}	6	8	9	12	16	18	24	32	36	48	72	96
3	1.000 ms	16	12	N/A	8	6	N/A	4	3	N/A	2	N/A	1
4	1.333 ms	12	9	8	6	N/A	4	3	N/A	2	N/A	1	N/A
6	2.000 ms	8	6	N/A	4	3	N/A	2	N/A	N/A	1	N/A	N/A
8	2.667 ms	6	N/A	4	3	N/A	2	N/A	N/A	1	N/A	N/A	N/A
9	3.000 ms	N/A	4	N/A	N/A	2	N/A	N/A	1	N/A	N/A	N/A	N/A
12	4.000 ms	4	3	N/A	2	N/A	N/A	1	N/A	N/A	N/A	N/A	N/A
24	8.000 ms	2	N/A	N/A	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

B2B – SELF TEST FEATURE

8

Figure 20-4: Overview of Block Synchronization Approach.



9

B2B – SELF TEST FEATURE

PREPOLL

Init

- Should Prepare payload before this event start
- Sts start as 0
- Tx should send late
- The whole sequence will be initialize before

Respond

- Receive and decrypt Payload
- Rx window open infinitely
- Rx window should open before
- The receive Sts should be for poll
- Doesn’t care if error
- This is the sync frame if sequence is lost

B2B – SELF TEST FEATURE

Poll – Respond - Final

Init **Respond**

Only need for timestamp should be send/receive correctly without much to do

B2B – SELF TEST FEATURE

Final Data

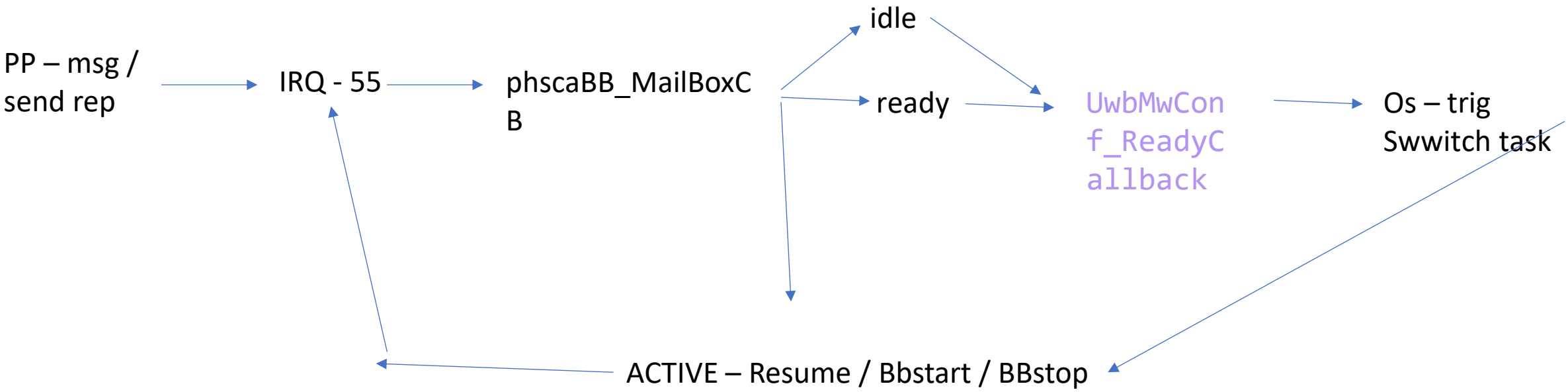
Init

- Prepare payload with all the timestamp
- After this should re trigger the whole sequence as it will not do on its own

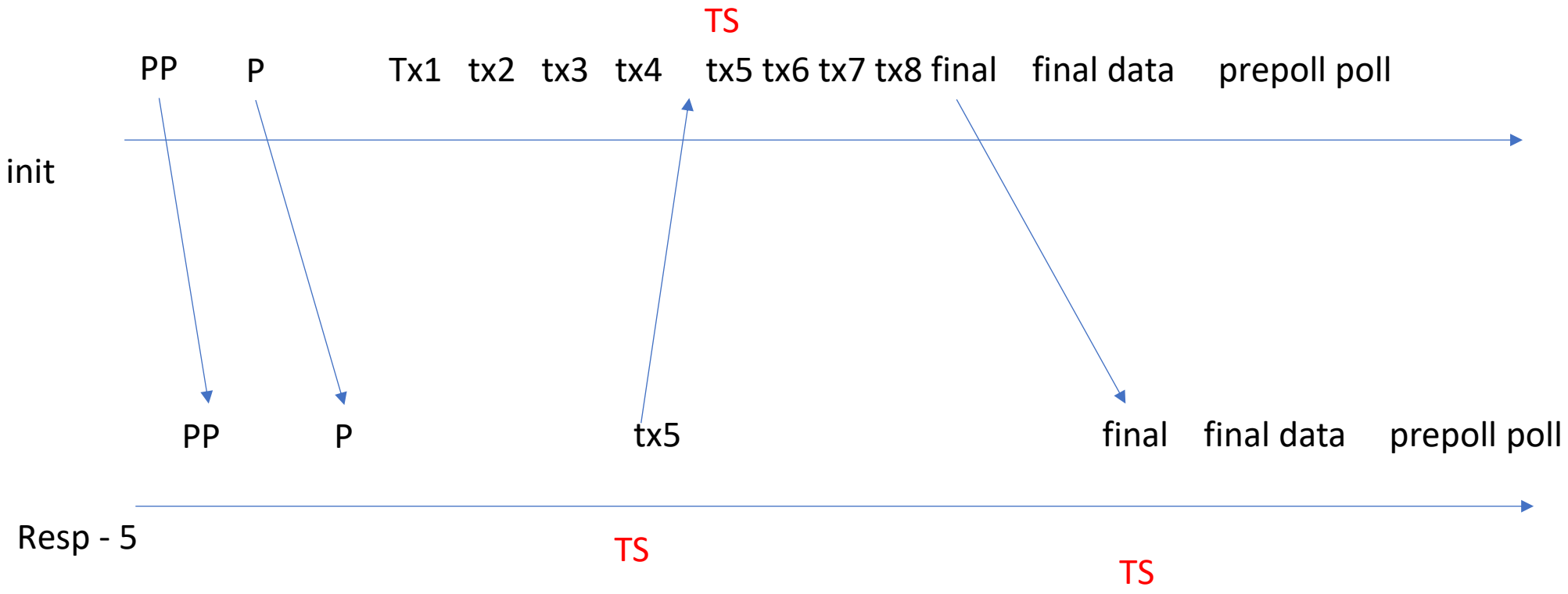
Respond

- Receive decrypt and send out the calculate Tof

CCC Ranging – actual sequence

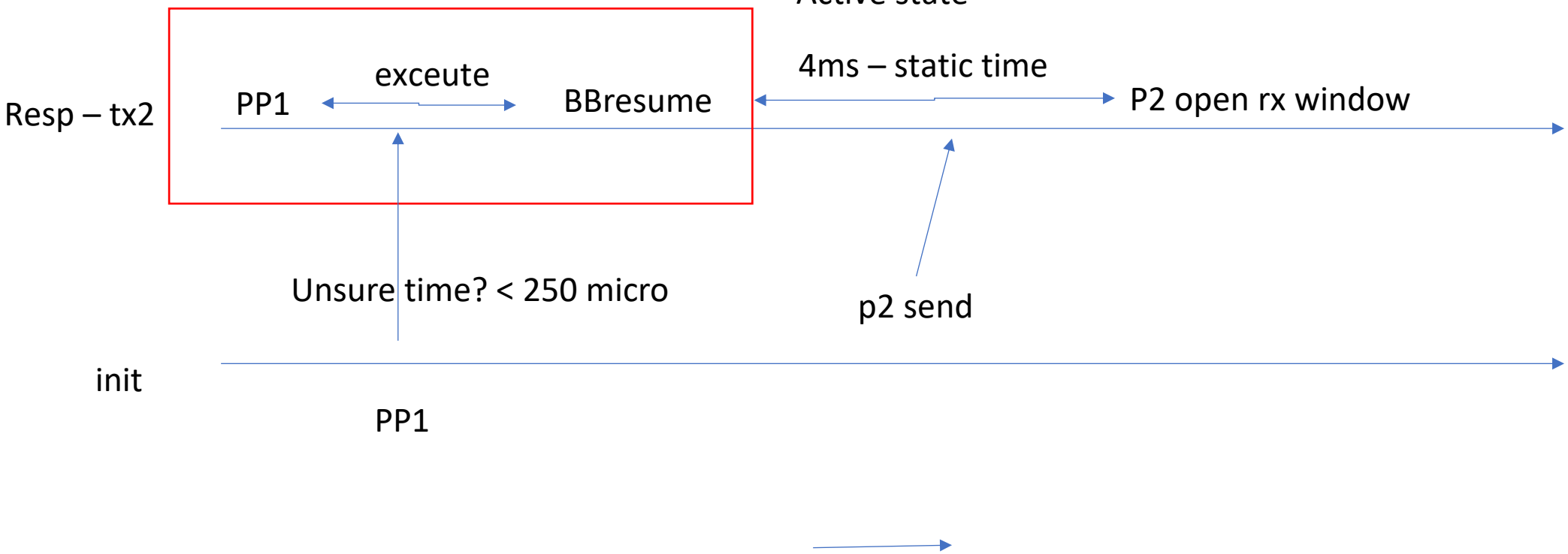


CALCULATE STS FOR THIS FRAME?

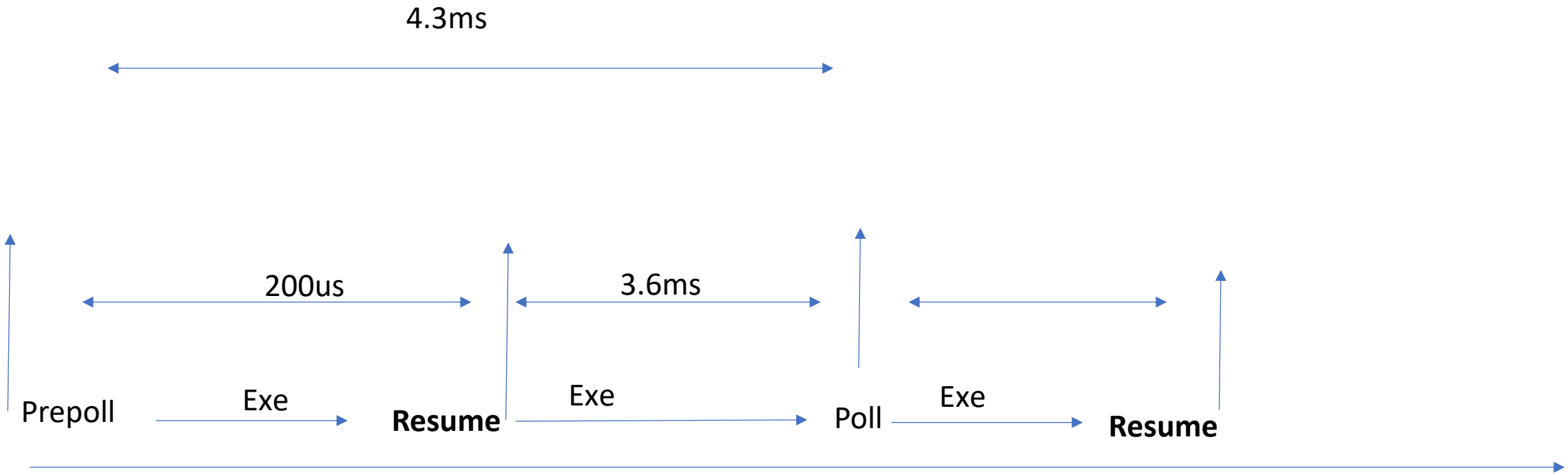


4,2ms -> 4,6ms

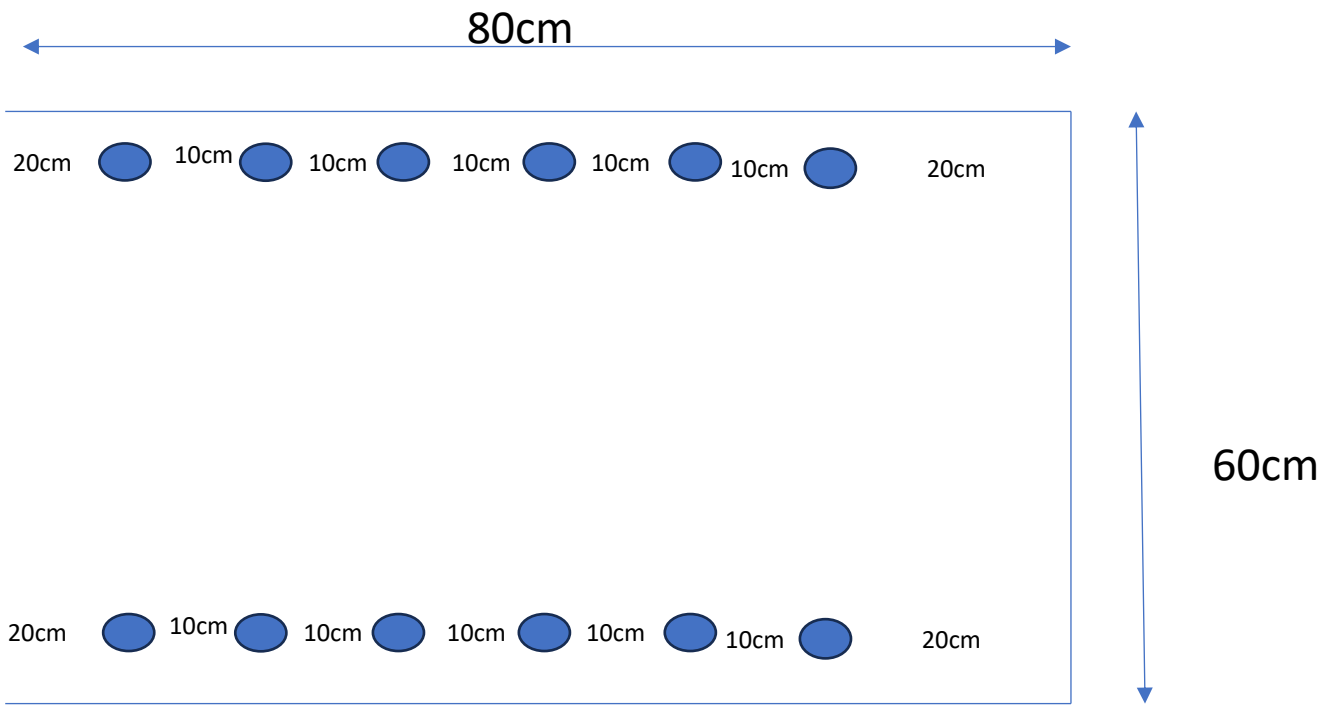
P2



10 -> 0 + 10 =
10 + 1



Get sts index :


80cm
20cm
10cm
10cm
10cm
10cm
10cm
20cm
60cm
20cm
10cm
10cm
10cm
10cm
10cm
20cm